Data warehousing and Big data

Week 8 and 9 - Topic 5 – Lesson 15, CLO5

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Outline

- Data warehouse and Big data
- OLAP
- star/snowflake schema
- ETL process

Data warehousing concept

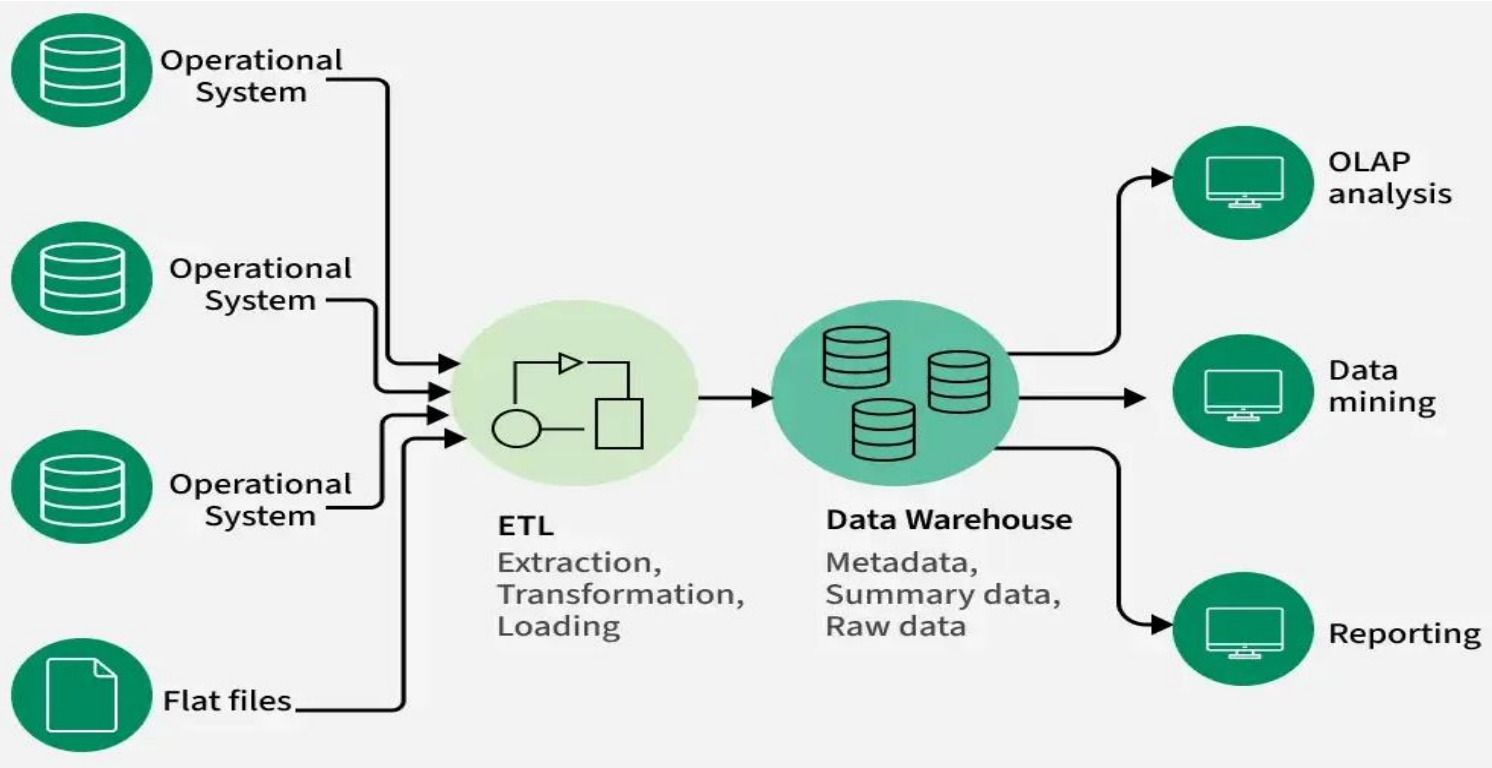
- Data warehousing is the process of collecting, integrating, storing and managing data from multiple sources in a central repository.
- It enables organizations to organize large volumes of current and historical data for efficient querying, analysis and reporting.

Need for Data Warehousing:

- Handling Large Data Volumes
- Enhanced Analytics
- Centralized Data Storage
- Trend Analysis
- Business Intelligence Support

Components of Data Warehouse

- Data Sources
- ETL (Extract, Transform, Load) Process
- Data Warehouse Database
- Metadata
- Data Marts
- OLAP (Online Analytical Processing) Tools
- End-User Access Tools



Types of Data Warehouses

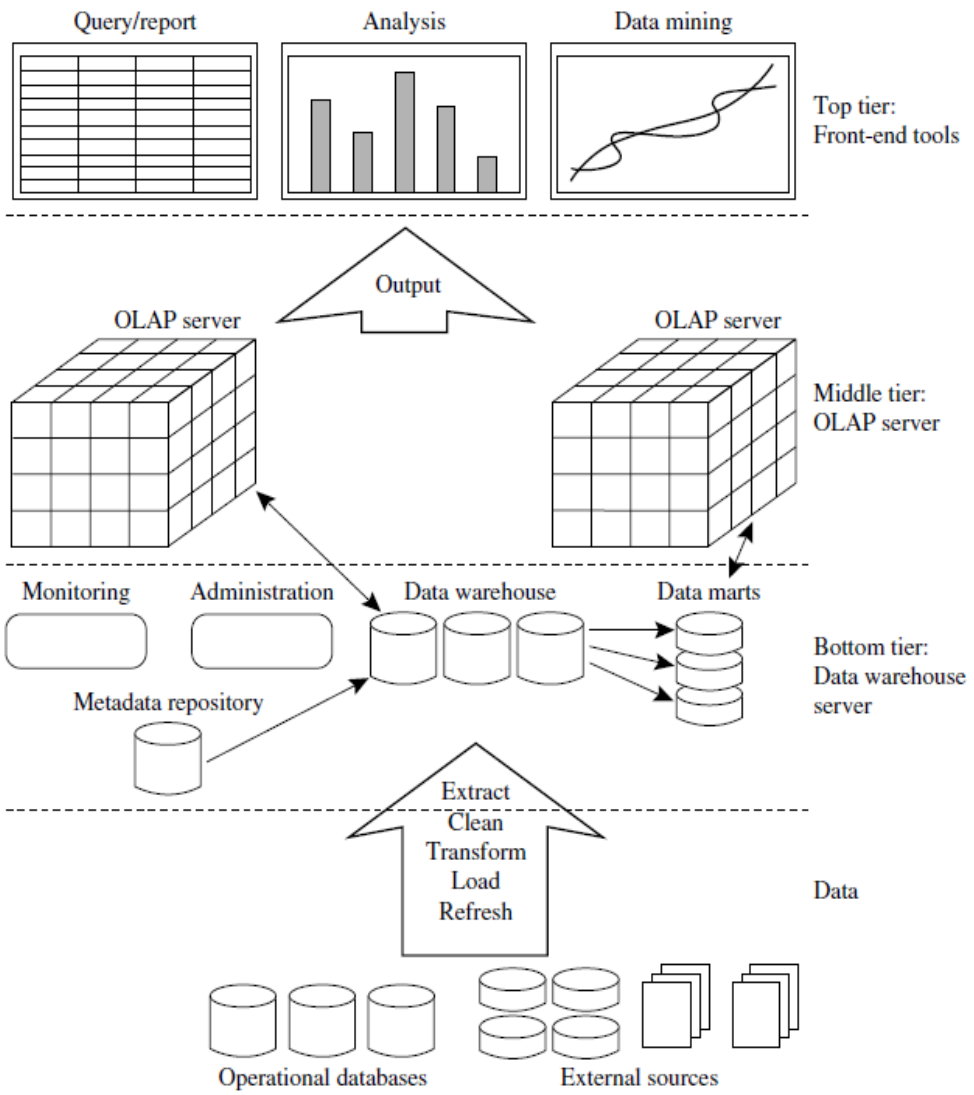
1. Enterprise Data Warehouse (EDW)
2. Operational Data Store (ODS)
3. Data Mart
4. Cloud Data Warehouse
5. Big Data Warehouse
6. Virtual Data Warehouse
7. Hybrid Data Warehouse
8. Real-time Data Warehouse

For example – A retail company collects data from:

- POS system (sales)
- Website (online orders)
- Customer database
- Inventory system

All this data is stored in a data warehouse for analysis such as:

- Monthly sales trends
- Best-selling products
- Customer purchasing behaviour

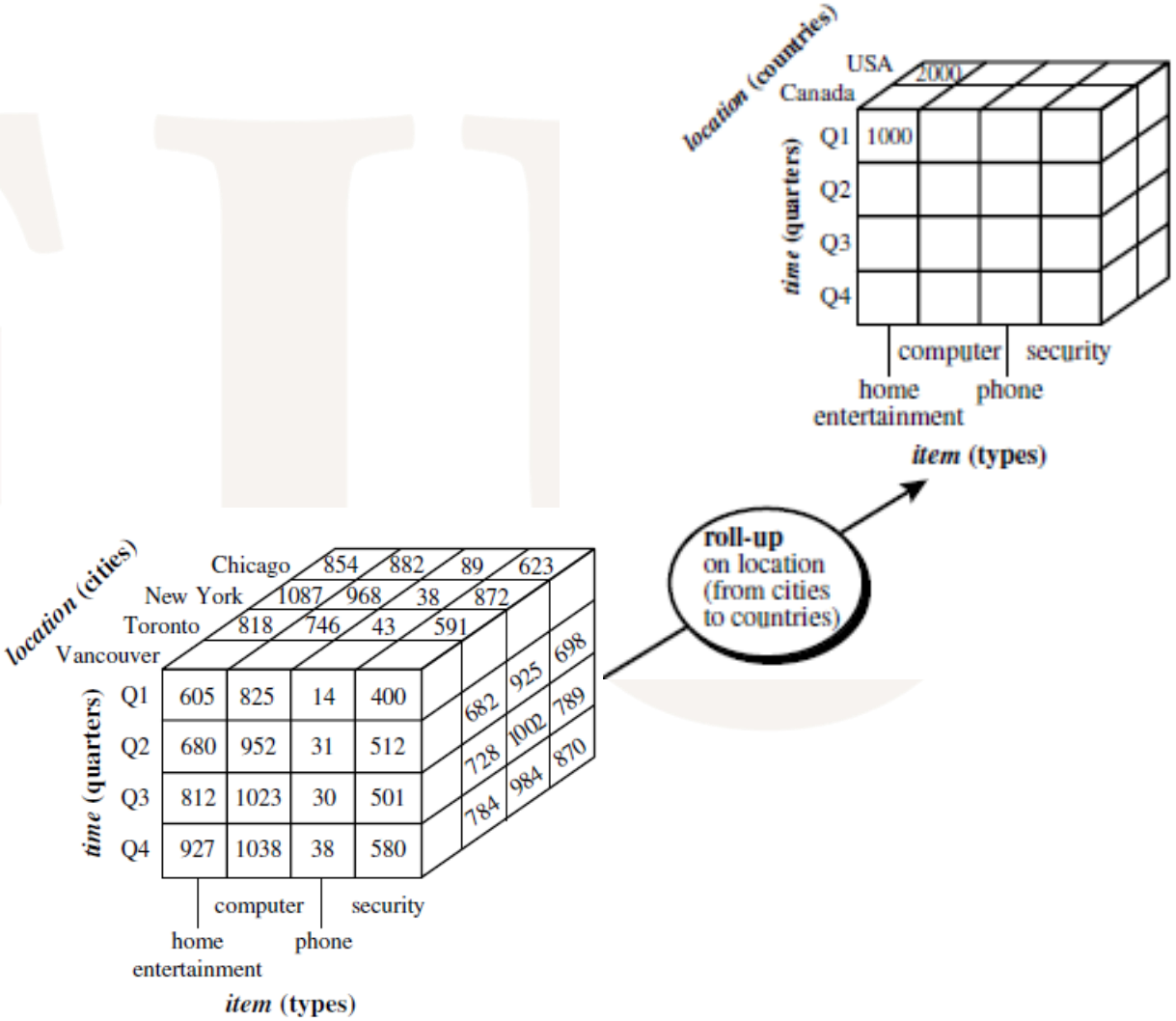


OLAP (Online Analytical Processing)

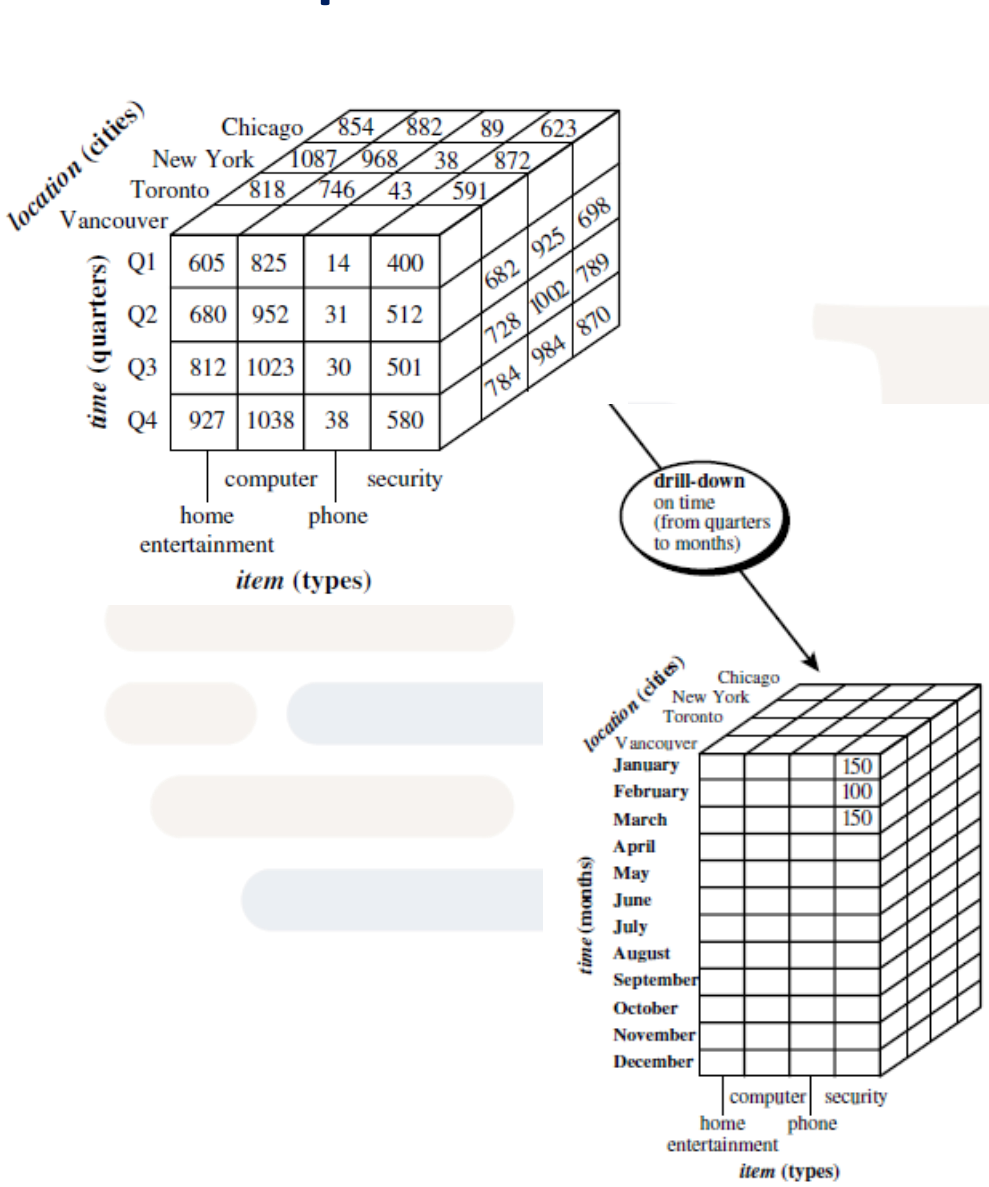
- OLAP is a technology used for fast multidimensional analysis of data stored in a data warehouse

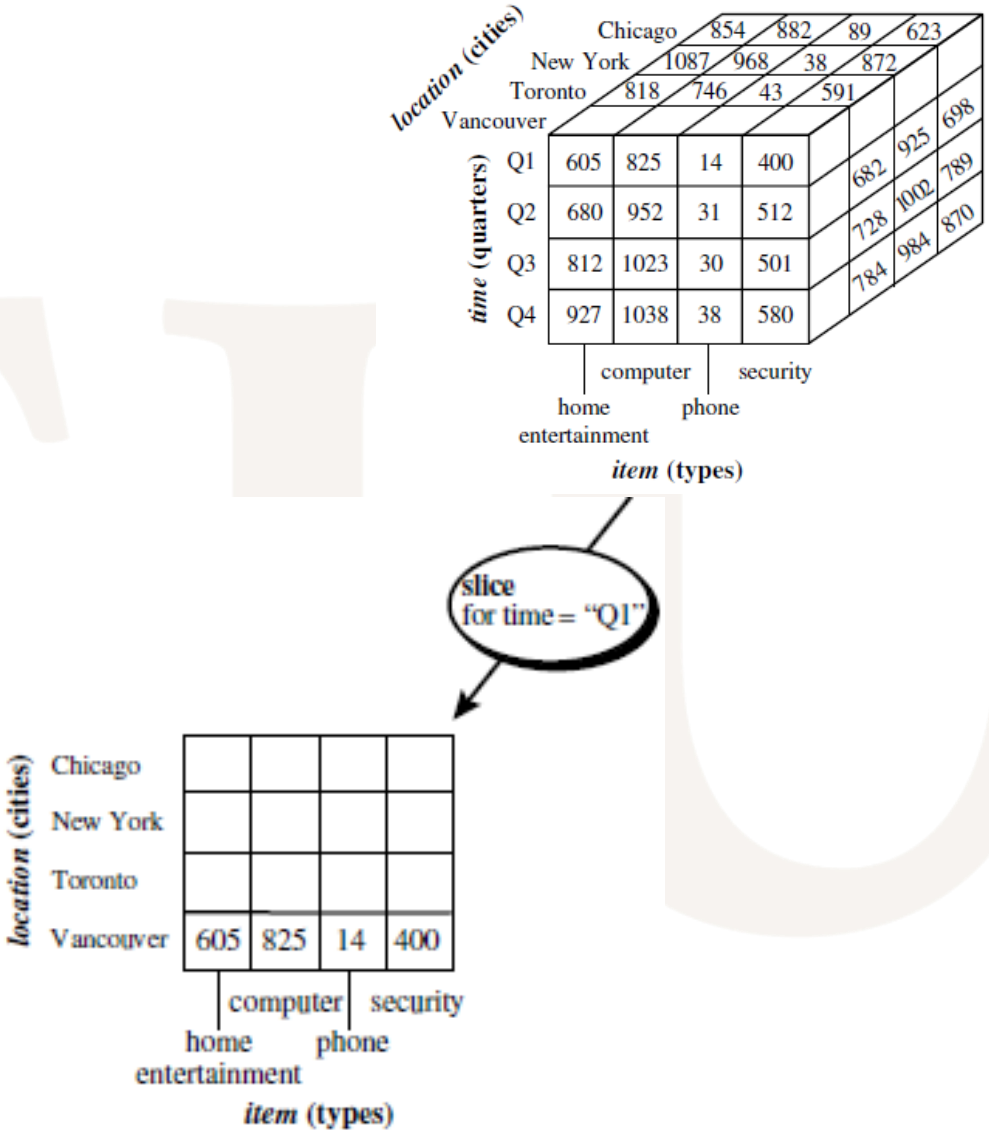
Types of OLAP operations –

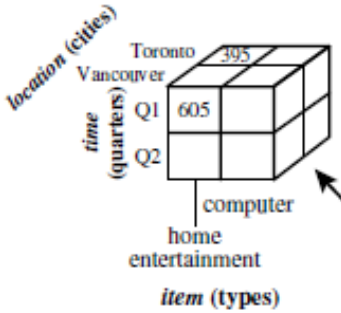
1. Roll up (drill-up): summarize data by climbing up hierarchy or by dimension reduction.
2. Drill down (roll down): reverse of roll-up from higher level summary to lower level summary or detailed data, or introducing new dimensions
3. Slice and dice: project and select
4. Pivot (rotate): reorient the cube, visualization, 3D to series of 2D planes



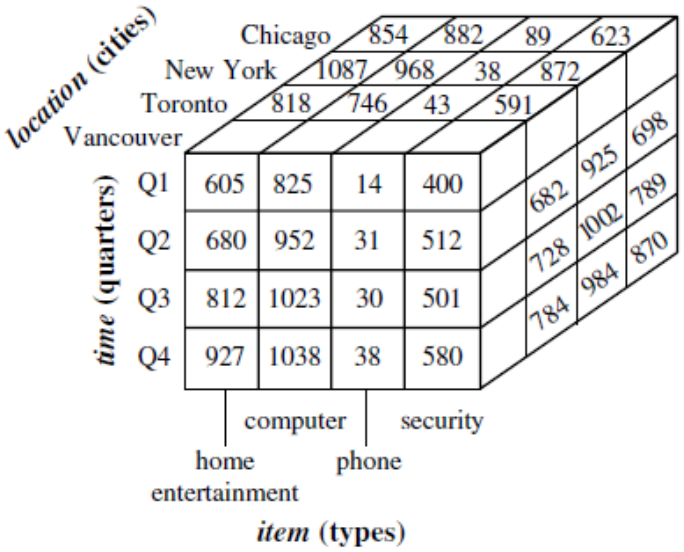
OLAP operations…

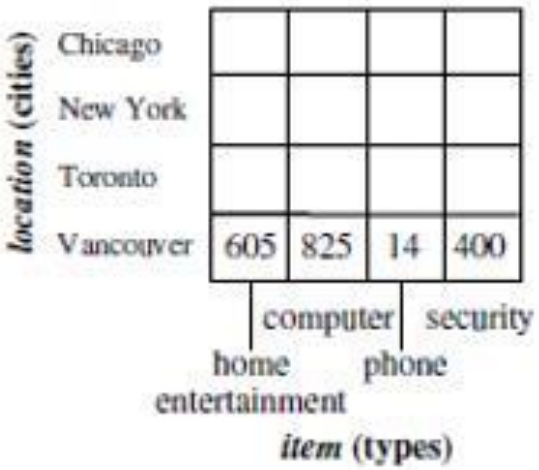


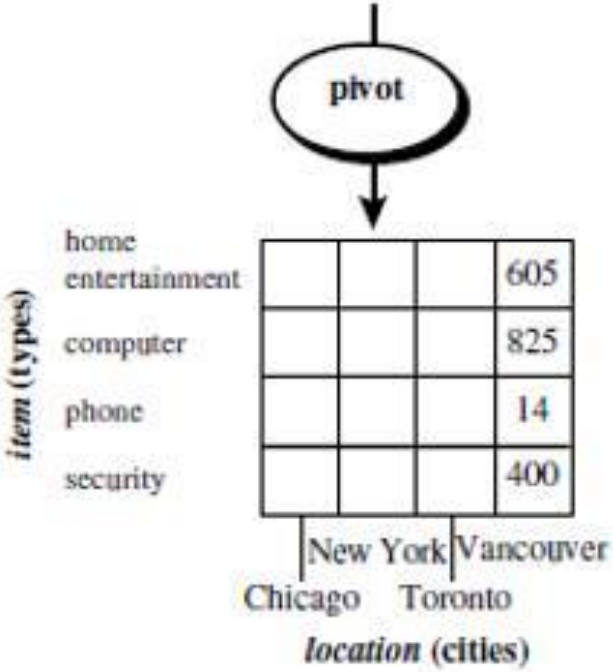




dice for
(location = "Toronto" or "Vancouver")
and (time = "Q1" or "Q2") and
(item = "home entertainment" or "computer")

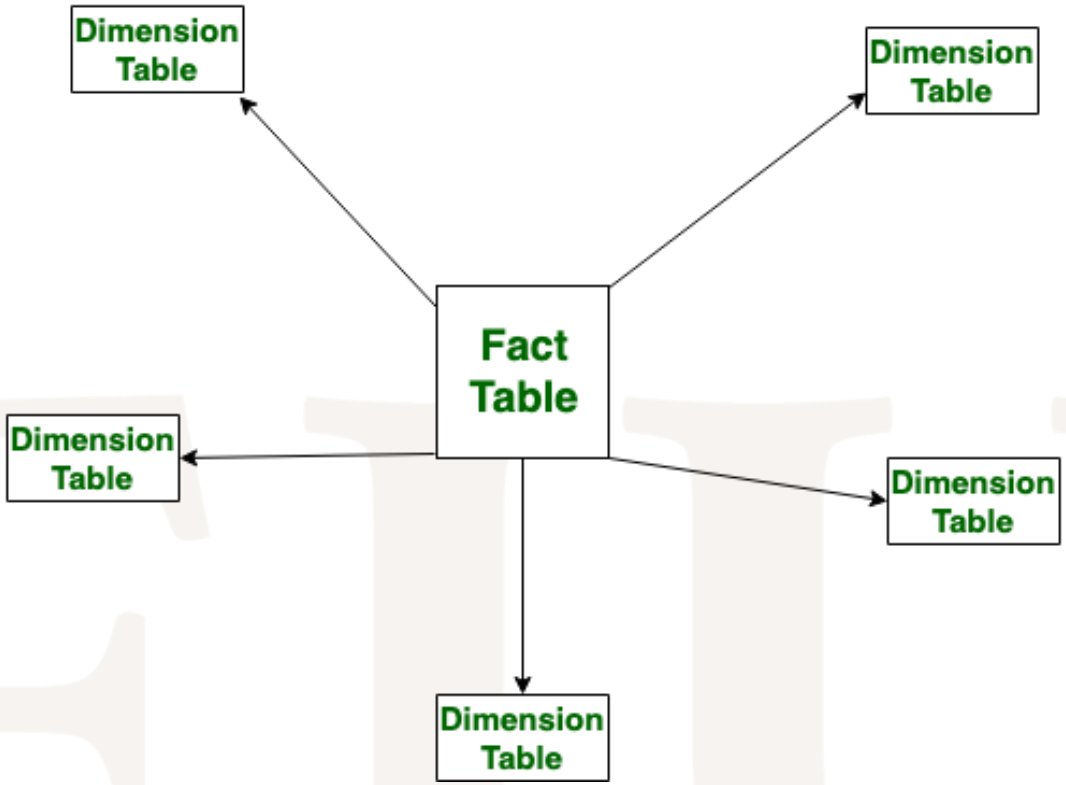






Star Schema

- Star Schema is a type of multidimensional model used for data warehouses.
- In a star schema, the fact tables and dimension tables are included.
- This schema uses fewer foreign-key joins.
- It forms a star structure with a central fact table connected to the surrounding dimension tables.



Example – Fact table

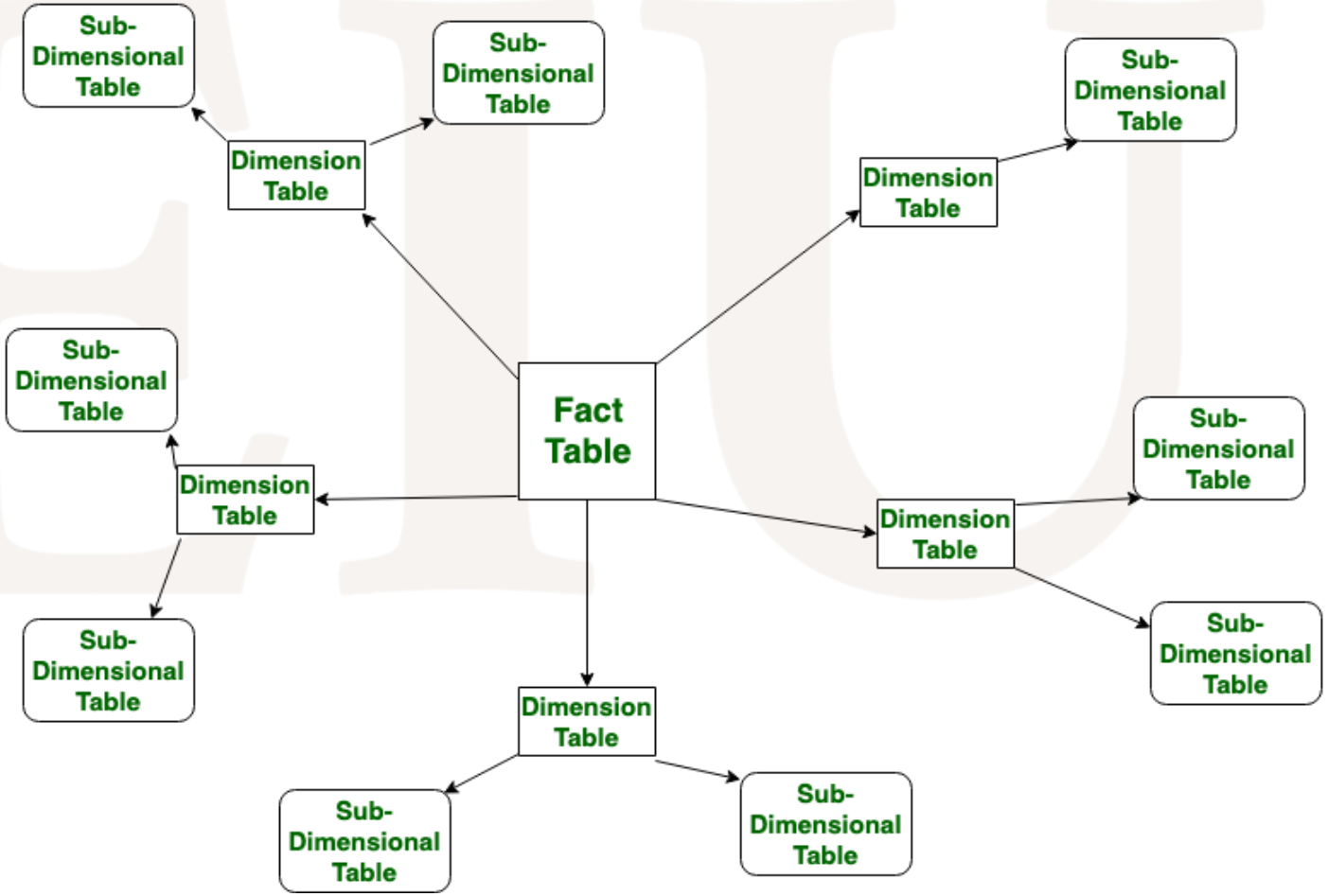
Product_ID	Customer_ID	Time_ID	Store_ID	Sales_Amount

Dimension Table –
Product
| Product_ID | Product_Name | Category |
Customer
| Customer_ID | Name | City |
Time
| Time_ID | Day | Month | Year |

Snowflake Schema

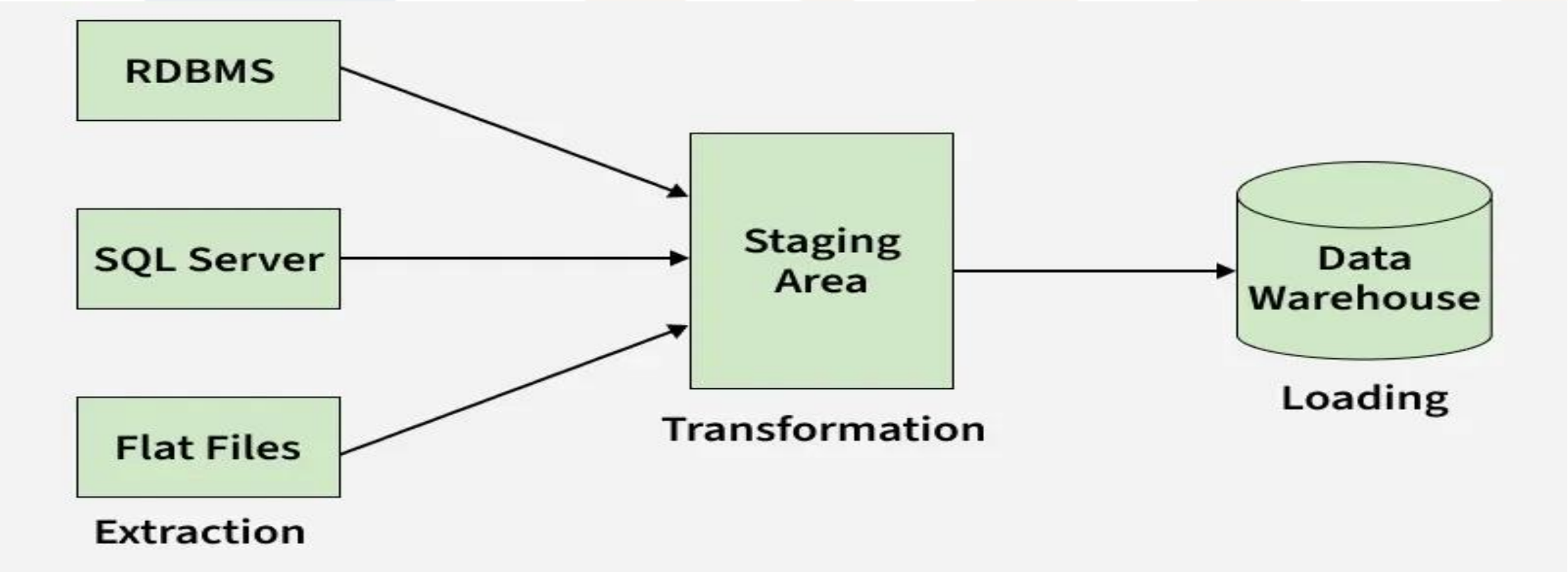
- In the snowflake schema, the fact tables, dimension tables and sub-dimension tables are included.
- This schema forms a snowflake structure with fact tables, dimension tables and sub-dimension tables.

Example –
Instead of storing category in Product:
Product Table
| Product_ID | Product_Name | Category_ID |
Category Table
| Category_ID | Category_Name |



ETL process

- The ETL process, which stands for Extract, Transform and Load, is a critical methodology used to prepare data for storage, analysis and reporting in a data warehouse.
- It involves three distinct stages that help to streamline raw data from multiple sources into a clean, structured and usable form.



Example:

Suppose a company has

- Excel sales file
- MySQL customer database
- Website order system

Step 1: Extract

Data pulled from: MySQL, Excel, Web database

Step 2: Transform

Data cleaning tasks: Convert currencies, Remove duplicates, Standardize date formats, Calculate total sales.

Raw Data:

Cust_ID | Price | Quantity

After Transformation:

Cust_ID | Total_Sales

101 | 200

Step 3: Load

Cleaned data is loaded into the data warehouse tables.

Why OLAP different from OLTP

- Online Transaction Processing (OLTP) focuses on handling large numbers of transactional operations in real time, ensuring data consistency and reliability for daily business operations.
- OLAP is designed for complex queries and data analysis, enabling businesses to derive insights from vast datasets through multidimensional analysis.

Further reading and practice

1. Silberschatz, A., Korth, H. F., & Sudarshan, S. (2020). Database System Concepts (7th ed.). McGraw-Hill. Ch.22 (pp. 881–900)